

ECSE-4040

Digital Electronics

Lab 1

Prof. T. Paul Chow

Bldg. CII, Room 6111

Rensselaer Polytechnic Institute

Troy, NY 12180-3590

Tel. (518)276-2910

FAX (518)276-2990

e-mail: chowt@rpi.edu

<https://lms.rpi.edu>

MOSFET Summary

Operating voltages for the NMOS and PMOS devices:

NMOS Voltages

$$V_{TN} \geq 0 \text{ V}$$

$$V_{GS} \geq 0 \text{ V}$$

$$V_{DS} \geq 0 \text{ V}$$

$$V_{BS} \leq 0 \text{ V} \text{ or } V_{SB} \geq 0 \text{ V}$$

PMOS Voltages

$$V_{TP} \leq 0 \text{ V} \text{ or } |V_{TP}| \geq 0 \text{ V}$$

$$V_{GS} \leq 0 \text{ V} \text{ or } |V_{GS}| \geq 0 \text{ V}$$

$$V_{DS} \leq 0 \text{ V} \text{ or } |V_{DS}| \geq 0 \text{ V}$$

$$V_{BS} \geq 0 \text{ V}$$

The NMOS threshold voltage can be computed as follows (with appropriate changes for PMOS device):

$$V_T = V_{T0} + \gamma (\sqrt{|2\phi_F| + V_{SB}} - \sqrt{|2\phi_F|})$$

$$V_{T0} = \phi_{GC} - 2\phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}} + \frac{Q_I}{C_{ox}}$$

$$\phi_F = \frac{kT}{q} \ln \frac{n_i}{p} \text{ (p-type substrate)}$$

$$\gamma = \frac{1}{C_{ox}} \sqrt{2q\epsilon_{si}N_A}$$

MOSFET Summary

For long-channel devices, the current equations are as follows:

If $V_{GS} \geq V_T$

if $V_{DS} \geq (V_{GS} - V_T)$ saturation region $I_{DS} = \frac{k}{2} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$

if $V_{DS} < (V_{GS} - V_T)$ linear region $I_{DS} = \frac{k}{2} [2(V_{GS} - V_T)V_{DS} - V_{DS}^2]$

If $V_{GS} < V_T$

subthreshold conduction region $I_{DS} = 0$

MOSFET Summary

For the velocity saturated short-channel devices, use

If $V_{GS} \geq V_T$

$$\begin{aligned} \text{if } V_{DS} \geq \frac{(V_{GS} - V_T)E_c L}{(V_{GS} - V_T) + E_c L} & \quad \text{saturation region} & I_{DS} = W v_{sat} C_{ox} \frac{(V_{GS} - V_T)^2}{(V_{GS} - V_T) + E_c L} \\ \text{if } V_{DS} < \frac{(V_{GS} - V_T)E_c L}{(V_{GS} - V_T) + E_c L} & \quad \text{linear region} & I_{DS} = \frac{W}{L} \frac{\mu_e C_{ox}}{\left(1 + \frac{V_{DS}}{E_c L}\right)} \left(V_{GS} - V_T - \frac{V_{DS}}{2}\right) V_{DS} \end{aligned}$$

If $V_{GS} < V_T$

$$\text{subthreshold conduction region} \quad I_{DS} = I_s e^{\frac{q(V_{GS} - V_T - V_{offset})}{nkT}} \left(1 - e^{\frac{-qV_{DS}}{kT}}\right)$$

Process Parameters

P2.1 Below is a table of process parameters for a generic 0.13 μm technology.

Table P2.1

0.13 μm Process technology parameters

Typical Technology Parameters	Symbol	NMOS	PMOS
Gate thickness	t_{ox} [Å]	22	22
Poly doping level	N_D [cm^{-3}]	3×10^{20}	—
Poly doping level	N_A [cm^{-3}]	—	3×10^{20}
Substrate doping level	N_A [cm^{-3}]	3×10^{17}	—
N-well doping level	N_D [cm^{-3}]	—	3×10^{17}
Number of surface state charges	N_f or N_{ss} [cm^{-2}]	6×10^{11}	6×10^{11}

Question Overview

HJS P2.1

- In this question you will practice calculating V_{T0} the threshold voltage of MOSFETs.
- Additionally, you will practice performing the calculations needed to pick a threshold voltage offset implant, allowing a desired V_{T0} to be obtained.

HJS P2.3

- Here, you will get familiar with the I-V characteristics of MOSFETs. You will determine whether the long-channel (quadratic) model holds for our choice of technology node.

HJS P2.5

- This question contains a NMOS with a capacitor in series to GND.
- This structure is the model of Dynamic Random-Access Memory (DRAM)
 - DRAM is used to store every bit of memory in your laptop, desktop, phone tablet, any computing device.
 - You are reading this lab assignment because your computer has billions of the circuits in this question.
- You will calculate by hand the expected V_X node voltage, which is the voltage that is stored in the memory cell.
- You will then verify this DC operating voltage in LTspice.

HJS P2.1 (6 Points)

Calculate the following values using the table in slide **L1-5**.

- a) Compute the flatband voltage (V_{FB}) and V_{T0} at zero bias for both NMOS and PMOS devices. Assume that there is no charge in the bulk oxide itself, but that there is a sheet charge layer at the Si-SiO₂ interface.
- b) Normally, NMOS gate polysilicon is heavily doped with donors (n^+), while PMOS gate polysilicon is heavily doped with acceptors (p^+).
 - i. How would V_{T0} be affected if the PMOS polysilicon gate was doped with donors rather than acceptors?
 - ii. Calculate this new V_{T0} .
- c) Calculate V_{T0} if Molybdenum ($\Phi_{G(gate)}=0$) is used for both the NMOS and PMOS gate electrodes instead of doped polysilicon.
- d) Adjust the threshold voltages of both the NMOS and PMOS devices so that the following threshold voltages are achieved: $V_{T0N}=0.3V$ and $V_{T0P}=-0.3V$. Calculate the threshold implant concentrations needed to achieve this for the cases in parts (a) and (b). Assume that the ion implantation depth is 25nm (not 100nm as in lecture notes).
- e) Why do modern technologies have an n^+ poly gate for NMOS and a p^+ gate for PMOS devices?

HJS P2.3 (6 Points)

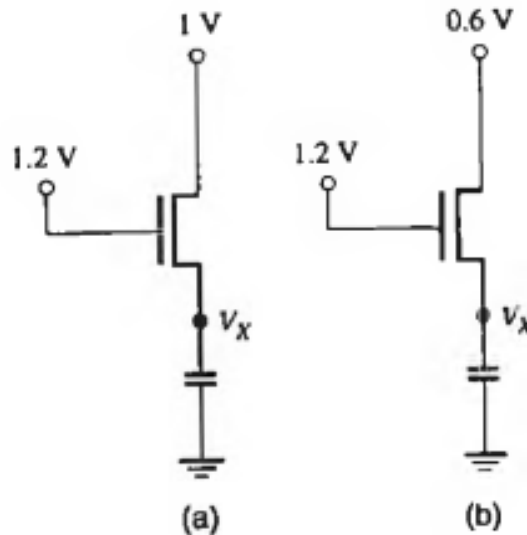
Calculate the following values using the table in slide **L1-2**.

- a) This problem involves plotting several I-V characteristics of NMOS and PMOS transistors in a $0.13\mu\text{m}$ technology. The $0.13\mu\text{m}$ technology uses a 1.2V power supply. The transistors are unit sized where $W = 100\text{nm}$, $L = 100\text{nm}$.
- Determine whether this MOSFET is a long or short channel transistor.
 - Plot I_{DS} vs. V_{DS} as a function of $V_{\text{GS}} = 0, 0.4, 0.8$ and 1.2V for both NMOS and PMOS devices.
 - Plot I_{DS} vs. V_{GS} with $V_{\text{DS}} = 1.2\text{V}$ for the NMOS device. Does the quadratic model hold for the $0.13\mu\text{m}$ devices, or is it closer to linear with V_{GS} (see HJS Fig. 2.11)?
 - Compare the hand calculations of i and ii with SPICE simulations.

HJS P2.5 (8 Points)

In the circuits in the figure below, determine the voltage across the capacitor (V_X) after the circuit reaches steady state. Assume the capacitor is initially discharged ($V_X=0$) and $V_{T0} = 0.4\text{V}$.

- Assume $L=130\text{nm}$, $W/L = 1$, $C = 10\text{pF}$
- Answer in two ways:
 - a) By explaining the region of operation of the MOSFET.
 - b) SPICE simulations.

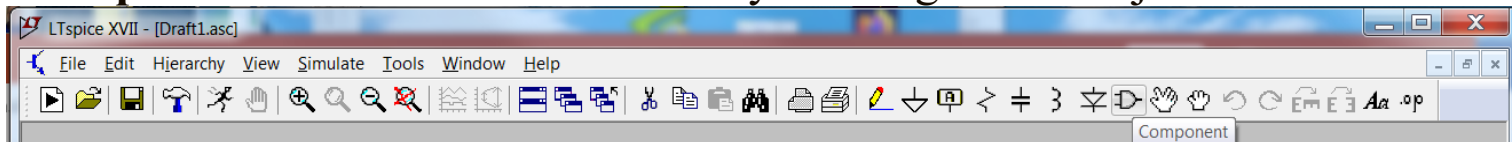


LTspice

- <https://www.analog.com/en/design-center/design-tools-and-calculators/ltspice-simulator.html>
 - Download the correct version for your operating system.
- Begin by creating a **New Schematic**.



- You will need to place instances of the following components:
 - nmos4 – NMOS transistor
 - voltage – DC voltage sources
 - capacitor
 - ground
 - **nmos4** and **voltage** are found by clicking the **Component** icon and browsing the components list.
 - **Capacitor** and **Ground** are found by clicking the icons just to the left of **Component**.



LTspice

- Transistor Length and Width are set by **right-click** of the NMOS transistor.
- V_{T0} and K' electrical parameters are set with **SPICE Directive** text.
 - This text can be added to the schematic by clicking the rightmost **.op** button in the top toolbar.
 - The following text will correctly set V_{T0} and K'
 - **.model NMOS nmos(kp=500u vt0=0.4)**
- Voltages and capacitance are set by **right-click** of the respective symbols.
- Wiring is performed with the **Wire** icon in the main toolbar with **Component**, **Capacitor**, etc.
- The V_X node can be labeled with the **Label Net** icon in the main toolbar.
 - After selecting a name and clicking **OK**, click the wire that you want to label.
- To simulate, click the **Run** icon in the main toolbar.
 - Select **DC op pnt**, then **OK** to run.

Report Submission

Submit your work on LMS under **Labs**

- Show your work.
 - Include images of your hand calculations with work and final results.
 - Include screenshots of Ltspice with the schematic you simulated and the V_X output voltage for both circuits.
- Due date: Friday at 11:59pm.